Rudder Angle Position Indicator – General data

1. Description

A potentiometer is mounted forward of the rudder hinge bracket assembly near station WL 50.15 and sends a signal of the rudder angle position through the DAU and onto the Flight Data Recorder (if installed).

2. Removal/Installation

Equipment and Materials:

- Corrosion preventative compound MIL–C–16173 Grade 2 or equivalent

A. Remove the Rudder Angle Position Potentiometer

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

3) Disconnect the electrical connection at the terminal block.

4) Loosen the nut and screw that hold the lever assembly to the potentiometer shaft mounted in mounting bracket. Remove lever assembly.

5) Remove the safety wire and the (3) screws and clamps which secure the potentiometer onto the mounting bracket.

6) Remove potentiometer.

B. Install the Rudder Angle Position Potentiometer

Note Coat all bolt shanks with corrosion preventative compound MIL–C–16173 Grade 2 or equivalent before installation.

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Ensure the FLIGHT CONTROLS INDICATION (A14) circuit breaker is pulled and fitted with safety clip.

3) Position the potentiometer on to the mounting bracket. Secure with the (3) clamps and screws. Safety wire screws.

4) Position the mounting bracket on channel and secure in place with (4) bolts, washers and nuts.

5) Locate the lever assembly on potentiometer shaft leaving a gap of 0.02 to 0.25 inch between the lever assembly and the potentiometer housing to minimize the push rod assembly misalignment.

6) Secure with a washer, nut and screw.

7) Install bolt through the push rod assembly end, install washer, and then bolt through the lever assembly end. Install washer, nut and cotter pin.

8) Adjust the rudder angle position potentiometer. (Refer to Para 3.A).

9) Reconnect the terminals 1, 2 and 3 to the electrical connection at terminal block.

10) Remove safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

11) Do a Rudder Position Voltage Check per Para 3.B.

3. Adjustment/Test

A. Adjustment Rudder Angle Position Potentiometer

Equipment and Material:

- DC Voltmeter

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Measure the Ground Power Voltage (V_{GP}) at the external power relay (or other suitable location).

3) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

4) Move the rudder to neutral and lock in place (refer to 27-20-00, Rudder Control System – General Data).

5) Disconnect terminal 2 of the potentiometer electrical connection at the terminal block and connect voltmeter positive to the lead from terminal 2 while attaching voltmeter negative to aircraft ground.

CAUTION **ENSURE THE TERMINAL 2 LEAD DOES NOT COME INTO CONTACT WITH AN ELECTRICAL GROUND.**

6) Loosen screw and nut on lever at the potentiometer shaft enough that the shaft turns freely.

7) Reset the FLIGHT CONTROLS INDICATION (A14) circuit breaker.

8) Rotate potentiometer shaft until measured DC Voltage is precisely 50% of total aircraft voltage V_{GP} (approximately 14 volts).

9) Reconnect pushrod and lever assembly to potentiometer shaft.

Note Do not rotate potentiometer shaft during this process.

10) Tighten screw and nut on lever leaving a gap of 0.02 to 0.05 between lever assembly and potentiometer housing.

11) Remove rudder control lock.

12) With voltmeter connected, function test rudder control position per the following:

a) Deflect rudder to the full left position (Reference 17 degrees) and verify measured voltage is 78.3% of V_{GP}.

b) Deflect rudder to the full right position (Reference 21 degrees) and verify measured voltage is 15.0% of V_{GP}.

Note Rudder may be configured for 25 degrees full left/right deflection. For 25 degrees full left, measured voltage is 91.7% of V_{GP} and for 25 degrees full right, measured voltage is 8.3% of V_{GP}. Adjust procedure per applicable configuration.

13) Pull FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit with safety clip. Disconnect voltmeter from potentiometer.

14) Reconnect the potentiometer electrical connections at the terminal block.

15) Remove safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

16) Deflect the rudder full left and ensure DAU Maintenance Status CAS message does not activate.

17) Deflect the rudder full right and ensure DAU Maintenance Status CAS message does not activate.

18) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Note For Flight Data Recorder input checks, refer to 31-31-00, Flight Data Recorder (FDR) – General Data.

For Data Acquisition Unit input checks, refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data.

B. Rudder Position Voltage Check

Equipment and Materials:

- DC Voltmeter

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Measure the Ground Power Voltage (V_{GP}) at the external power relay (or other suitable location).

3) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

4) Move the rudder to neutral and lock in place (refer to 27-20-00, Rudder Control System – General Data).

5) Disconnect terminal 2 of the potentiometer electrical connection at the terminal block and connect voltmeter positive to the lead from terminal 2 while attaching voltmeter negative to aircraft ground.

CAUTION ENSURE THE TERMINAL 2 LEAD DOES NOT COME INTO CONTACT WITH AN ELECTRICAL GROUND.

6) Reset the FLIGHT CONTROLS INDICATION (A14) circuit breaker and monitor DC voltage from terminal 2 to aircraft ground.

7) If DC Voltage is not 50% of V_{GP}, do adjustment steps per Para 3.A.

8) Remove rudder control locks.

9) With voltmeter connected, function test rudder control position per the following:

a) Deflect rudder to the full left position (Reference 17 degrees) and verify measured voltage is 78.3% of V_{GP}.

b) Deflect rudder to the full right position (Reference 21 degrees) and verify measured voltage is 15.0% of V_{GP}.

Note Rudder may be configured for 25 degrees full left/right deflection. For 25 degrees full left, measured voltage is 91.7% of V_{GP} and for 25 degrees full right, measured voltage is 8.3% of V_{GP}. Adjust procedure per applicable configuration.

10) If DC Voltage does not correspond, do adjustment steps per Para 3.A.

11) Pull FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit with safety clip.

12) Disconnect voltmeter from potentiometer.

13) Reconnect the potentiometer electrical connections at the terminal block.

14) Remove safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

15) Deflect the rudder full left and ensure DAU Maintenance Status CAS message does not activate.

16) Deflect the rudder full right and ensure DAU Maintenance Status CAS message does not activate.

17) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Note For Flight Data Recorder input checks, refer to 31-31-00, Flight Data Recorder (FDR) – General Data.

For Data Acquisition Unit input checks, refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data.